

Supplementary Material: Improving Steady-State Precision of Reset Controllers via Integral Sliding Mode Control

Xinxin Zhang, *Member, IEEE*, and Leonid Freidovich, *Senior Member, IEEE*

I. STABILITY VERIFICATION OF THE RESET CONTROL SYSTEMS

A. Preliminaries

Definition 1 (Closed-loop reset control system [1] in the manuscript). Consider a closed-loop reset system $\mathcal{H}_\chi$ comprising the plant and the reset controller, interconnected via the control law $u(t) = u_r(t)$ and satisfying Assumptions 1-3. Defining the closed-loop state $\chi(t) = [e_s^\top(t), x_r^\top(t)]^\top \in \mathbb{R}^{n+n_r}$, the dynamics of $\mathcal{H}_\chi$ are given by

$$\begin{cases} \dot{\chi}(t) = A_\chi \chi(t) + E_r q_r(t) + B_c d(t), & \text{if } \zeta(t) \in \mathcal{F}_r, \\ \chi(t^+) = A_{\chi,\rho} \chi(t), & \text{if } \zeta(t) \in \mathcal{J}_r, \end{cases} \quad (1)$$

where

$$\begin{aligned} A_\chi &= \begin{bmatrix} A - BD_R C & -BC_R \\ B_R C & A_R \end{bmatrix} \in \mathbb{R}^{(n+n_r) \times (n+n_r)}, \\ E_r &= \begin{bmatrix} I_n & 0_{n \times n_r} \end{bmatrix}^\top \in \mathbb{R}^{(n+n_r) \times n}, \\ B_c &= \begin{bmatrix} -B^\top & 0_{1 \times n_r} \end{bmatrix}^\top \in \mathbb{R}^{(n+n_r) \times 1}, \\ A_{\chi,\rho} &= \text{diag}(I_n, A_\rho) \in \mathbb{R}^{(n+n_r) \times (n+n_r)}. \end{aligned} \quad (2)$$

Assumption 5 (in the manuscript). For the closed-loop reset system (1), there exist a symmetric positive-definite matrix $P \in \mathbb{R}^{(n+n_r) \times (n+n_r)}$ and a constant $\lambda_0 > 0$ such that the flow condition

$$\chi^\top (A_\chi^\top P + P A_\chi) \chi \leq -\lambda_0 \chi^\top P \chi, \quad \forall \zeta(t) \in \mathcal{F}_r, \quad (3)$$

and the jump condition

$$\chi^\top (A_{\chi,\rho}^\top P A_{\chi,\rho} - P) \chi \leq 0, \quad \forall \zeta(t) \in \mathcal{J}_r, \quad (4)$$

are satisfied.

The $\mathcal{H}_\beta$ -condition [1], [2] provides a sufficient criterion to verify Assumption 5. Let the controller state be partitioned into $n_\rho = 1$ reset and $n_{\bar{\rho}} = n_r - n_\rho$ non-reset dimensions. The condition requires the fictitious transfer function

$$H_\beta(s) = C_\beta (sI - A_\chi)^{-1} B_\rho \quad (5)$$

Xinxin Zhang (e-mail: xinxin.zhang@umu.se) and Leonid Freidovich (e-mail: leonid.freidovich@umu.se) are with Department of Applied Physics and Electronics, Umeå University, Håken Gullerussons väg 20, Teknikhuset, 901 87 Umeå. This work is supported by the Kempe Foundation, Sweden, under project number JCSMK24-523.

to be SPR for some vector $\beta \in \mathbb{R}^{n_\rho}$ and symmetric matrix $P_\rho > 0$, where the extraction and injection matrices are defined as:

$$C_\beta = \begin{bmatrix} \beta C & P_\rho & 0_{n_\rho \times n_{\bar{\rho}}} \end{bmatrix}, \quad B_\rho = \begin{bmatrix} 0_{n_\rho \times n} & I_{n_\rho} & 0_{n_\rho \times n_{\bar{\rho}}} \end{bmatrix}^\top. \quad (6)$$

According to the Kalman-Yakubovich-Popov (KYP) Lemma [3, Lemma 6.3], the transfer function $H_\beta(s)$ is SPR if and only if A_χ is Hurwitz, the frequency response satisfies $\text{Re}[H_\beta(j\omega)] > 0$ for all ω , and the high-frequency limit evaluates to $\lim_{\omega \rightarrow \infty} \omega^2 \text{Re}[H_\beta(j\omega)] = -c_2 > 0$, where $c_2 = C_\beta A_\chi B_\rho$. When these frequency-domain requirements are satisfied, the existence of a positive-definite matrix $P > 0$ that fulfills the continuous flow condition $Q = -(A_\chi^\top P + P A_\chi) > 0$ while enforcing $P B_\rho = C_\beta^\top$ is ensured.

In this manuscript, we apply this theoretical framework to both the reset control systems.

B. Continuous Flow Dynamics and Spectrum Verification

We first verify that the unforced continuous flow dynamics of the base reset architectures are stable.

1) *PCI Control System*: The proportional-Clegg integrator (PCI) control system combines a PCI controller with a first-order linear time-invariant (LTI) plant:

$$A = -0.962, \quad B = 178.846, \quad C = 1, \quad D = 0. \quad (7)$$

The PCI controller is defined by:

$$A_R = 0, \quad B_R = 1, \quad C_R = 0.183, \quad D_R = 0.03. \quad (8)$$

For the state vector $\chi \in \mathbb{R}^2$, the closed-loop continuous flow matrix evaluates to:

$$A_{\chi,\text{PCI}} = \begin{bmatrix} A - BD_R C & -BC_R \\ B_R C & A_R \end{bmatrix} = \begin{bmatrix} -6.3274 & -32.7288 \\ 1 & 0 \end{bmatrix}. \quad (9)$$

Solving $\det(sI - A_{\chi,\text{PCI}}) = 0$ verifies the Hurwitz spectrum $\text{spec}(A_{\chi,\text{PCI}}) = \{-3.1637 \pm 4.7665i\}$.

2) *PCID Control System*: The proportional-Clegg integrator-derivative (PCID) architecture cascades a derivative filter in series with the plant, forming a third-order extended LTI plant. To align with the numerical implementation, the

proportional gain $k_p = 0.10$ is absorbed into the measurement matrix C . The extended plant matrices are defined as:

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -0.9620 & -1686.2946 \\ 0 & 0 & -25.1300 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 715.6688 \\ 8.0000 \end{bmatrix},$$

$$C = \begin{bmatrix} 0.10 & 0 & 0 \end{bmatrix}, \quad D = 0. \quad (10)$$

Note that this system structure format is only used for the ISS verification.

The PCI controller is defined by matrices:

$$A_R = 0, \quad B_R = 1, \quad C_R = 2.04, \quad D_R = 1. \quad (11)$$

Applying the established block structure, the closed-loop continuous flow matrix for the state vector $\chi \in \mathbb{R}^4$ evaluates to:

$$A_{\chi, \text{PCID}} = \begin{bmatrix} A - BD_R C & -BC_R \\ B_R C & A_R \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 & 0 & 0 \\ -71.5669 & -0.9620 & -1686.2946 & -1459.9643 \\ -0.8000 & 0 & -25.1300 & -16.3200 \\ 0.1000 & 0 & 0 & 0 \end{bmatrix}. \quad (12)$$

Solving $\det(sI - A_{\chi, \text{PCID}}) = 0$ yields the Hurwitz spectrum $\text{spec}(A_{\chi, \text{PCID}}) = \{-22.9774, -1.8282, -0.6432 \pm 4.6273i\}$.

Because the eigenvalues of $A_{\chi, \text{PCID}}$ and $A_{\chi, \text{PCI}}$ possess negative real parts, both architectures dissipate energy during continuous flow. This characteristic guarantees a baseline decay rate $\lambda_0 > 0$ for the continuous dynamics, which is established before imposing a shared Lyapunov matrix P constrained by the discrete jump condition.

C. Verification of the $\mathcal{H}_\beta$ Condition

1) *PCI Control System:* For the PCI system ($n = 1, n_\rho = 1, n_{\bar{\rho}} = 0$), the injection matrix simplifies to $B_\rho = [0, 1]^\top$. Applying the tuning parameter $\beta = 0.15$, the matrix $C = 1$, and $P_\rho = 1$ yields $C_\beta = \begin{bmatrix} 0.15 & 1 \end{bmatrix}$. This parametrization evaluates to $c_2 = -4.91$, satisfying the high-frequency properness requirement ($-c_2 > 0$). The Lyapunov matrices verifying this certificate evaluate to:

$$P = \begin{bmatrix} 0.0465 & 0.15 \\ 0.15 & 1 \end{bmatrix}, \quad Q = \begin{bmatrix} 0.2887 & 1.4716 \\ 1.4716 & 9.8186 \end{bmatrix}. \quad (13)$$

Because $P > 0$ and $Q > 0$, the continuous flow condition is certified. Refining the tuning to $\beta = 0.0967$ produces the Lyapunov matrix $P = \begin{bmatrix} 0.03055 & 0.09666 \\ 0.09666 & 1 \end{bmatrix}$, establishing a safe baseline continuous decay rate of $\lambda_0 = 6.32$.

2) *PCID Control System:* For the PCID architecture ($n = 3, n_\rho = 1, n_{\bar{\rho}} = 0$), the reset injection matrix maps to the fourth state vector coordinate, yielding $B_\rho = \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix}^\top$. To satisfy the high-frequency properness requirement, an approximation ($A_R = -10^{-4}$) is introduced to the reset integrator. Consequently, the CI in the PCID is approximately formulated as a First-Order Reset Element (FORE) utilizing

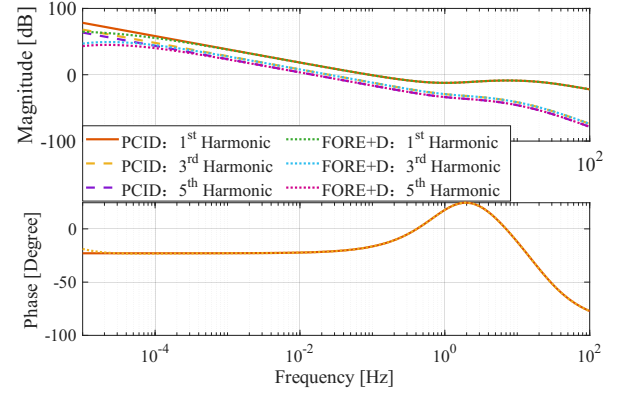


Fig. 1: Frequency response comparison of the first, third, and fifth harmonics for the ideal PCID controller and the approximated PCID controller ($A_R = -10^{-4}$), validating the FORE approximation.

a base linear controller (BLC) of $\frac{1}{s+10^{-4}}, s = j\omega$. As shown in Fig. 1, this FORE-based structure closely approximates the ideal PCID for frequencies $\omega > 10^{-4}$ rad/s. In practice, this approximation introduces a 10^4 s time constant, rendering its deviation negligible on experimental timescales. It is accommodated in the discrete implementation via the digital pole $z = e^{-10^{-6}} \approx 1 - 10^{-6}$ operating at a sampling time of $T_s = 0.01$ s.

Applying the tuning parameter $\beta = 0.45$ alongside $P_\rho = 1$, and utilizing $C = \begin{bmatrix} 0.10 & 0 & 0 \end{bmatrix}$, yields the extraction matrix $C_\beta = \begin{bmatrix} \beta C & P_\rho \end{bmatrix} = \begin{bmatrix} 0.0450 & 0 & 0 & 1 \end{bmatrix}$. Under this formulation, the high-frequency parameter evaluates to $-c_2 = 10^{-4} > 0$. The Lyapunov matrix verifying this structural certificate is:

$$P = \begin{bmatrix} 0.0026 & 0.0001 & -0.0027 & 0.0450 \\ 0.0001 & 0.0002 & -0.0141 & 0 \\ -0.0027 & -0.0141 & 1.2589 & 0 \\ 0.0450 & 0 & 0 & 1.0000 \end{bmatrix}. \quad (14)$$

This structural equality $PB_\rho = C_\beta^\top$ confirms the KYP requirement, and the matrix is positive definite. Evaluating the continuous flow condition $Q = -(A_{\chi, \text{PCID}}^\top P + PA_{\chi, \text{PCID}})$ with a targeted decay rate $\lambda_0 = 1.8 \times 10^{-4}$ verifies a strict positive-definite dissipation margin, establishing $\lambda_{\min}(Q - \lambda_0 P) \geq 1.25 \times 10^{-5}$.

While the ultimate Lyapunov tracking bounds derived in the manuscript depend on the matrix P and the decay rate λ_0 , these parameters are utilized here primarily as tools to prove closed-loop stability, rather than as predictive numerical metrics for practical experimental precision.

D. Discrete Energy Non-Expansion (Jump Dynamics)

For both systems, the KYP lemma structures the Lyapunov matrix to satisfy the equality $PB_\rho = C_\beta^\top$. Because resets are triggered within the reset set $\mathcal{J}_r$ (which is defined as a set where $e = 0$, rather than merely a surface), the cross-term $\beta C e$ evaluates to zero during jumps. With this cross-term vanishing, the discrete jump constraint simplifies to $(\gamma^2 - 1)P_\rho \leq 0$. Applying the nominal reset ratio $\gamma = -0.3$ and $P_\rho = 1$

yields $(\gamma^2 - 1)P_\rho = -0.91 < 0$. Combined with the verified continuous flow conditions, these baseline systems satisfy Assumption 5.

E. Stability of the RC-ISM and RC- Λ -ISM Systems

ISS for the PCI- Λ -ISM architecture is proven by combining the baseline PCI ISS with the ISM sliding reachability requirement. Enforcing the boundary layer constraint $\rho/\delta \geq \lambda_0/3$, utilizing the decay rate $\lambda_0 = 6.32$, fulfills Proposition 1 in the manuscript to verify that the PCI- Λ -ISM system is practically ISS (pISS) and uniformly ultimately bounded (UUB).

For the PCID-ISM system, fulfilling the baseline PCID ISS alongside the sliding reachability condition ensures the closed-loop system is ISS and UUB, as governed by Theorem 1 in [1].

II. TABLE OF NOTATION

TABLE I: Summary of notation used throughout the paper.

Symbol	Description
<i>Plant and Reference</i>	
$x(t)$	Plant state vector, $x(t) \in \mathbb{R}^n$
$u(t)$	Control input applied to the plant
$y(t)$	Plant output
$d(t)$	Matched disturbance
A, B, C	Plant state-space matrices; (A, B) controllable
n	Plant state dimension
$r_s(t)$	State reference trajectory
$r(t)$	Output reference, $r(t) = Cr_s(t)$
$e_s(t)$	State tracking error, $e_s(t) = r_s(t) - x(t)$
$e(t)$	Output tracking error, $e(t) = Ce_s(t)$
$q_r(t)$	Reference-feedforward related term, $q_r(t) = \dot{r}_s(t) - Ar_s(t)$
$\bar{r}_0, \bar{r}_1$	Bounds on $\ r_s(t)\ $ and $\ \dot{r}_s(t)\ $
$\bar{q}_r$	Bound on $\ q_r(t)\ $
Δ, Δ_1	Bounds on $ d(t) $ and $ \dot{d}(t) $
T_0	Fixed sampling period ($= 0.01$ s)
<i>Reset Control (RC) $\mathcal{C}_r$</i>	
$x_r(t)$	Reset controller state, $x_r(t) \in \mathbb{R}^{n_r}$
$u_r(t)$	Reset controller output
n_r	Reset controller state dimension
A_R, B_R, C_R, D_R	Reset controller continuous-flow matrices
A_ρ	Reset (jump) matrix, $A_\rho = \text{diag}(\gamma, I_{n_r-1})$
γ	Reset fraction, $\gamma \in (-1, 1)$
$\zeta(t)$	Triggering variable, $\zeta(t) = [x_r^\top(t), e(t)]^\top$
M	Symmetric matrix defining the reset boundary
$\mathcal{F}_r, \mathcal{J}_r$	Flow set and jump set of $\mathcal{C}_r$
t_k	Reset instant
<i>Closed-Loop Reset Control System $\mathcal{H}_\chi$</i>	
$\chi(t)$	Closed-loop state, $\chi(t) = [e_s^\top(t), x_r^\top(t)]^\top$
A_χ	Closed-loop flow matrix
$A_{\chi, \rho}$	Closed-loop reset (jump) matrix
E_r, B_c	Input matrices for $q_r(t)$ and $d(t)$ in χ -dynamics
P, λ_0	Lyapunov matrix and decay-rate constant (Assumption 4)
λ	Reduced decay rate, $\lambda = \lambda_0/3$
$V_e(\chi)$	ISS-Lyapunov candidate, $V_e(\chi) = \chi^\top P \chi$
σ_r, σ_d	ISS gains associated with $q_r(t)$ and $d(t)$
<i>Integral Sliding Mode Control (ISMC) $\mathcal{C}_s$</i>	
$s(t)$	Sliding variable
C_m	Sliding surface vector, $C_m \in \mathbb{R}^{1 \times n}$
G	Sliding gain, $G = C_mB \neq 0$

continued on next page

TABLE I – *continued from previous page*

Symbol	Description
$u_{sm}(t)$	ISMC control input
ρ	Switching gain
η	Reachability margin
δ	Boundary-layer thickness
$\text{sat}_\delta(\cdot)$	Saturation function with boundary layer δ
Ω_δ	Boundary layer set, $\{s : s \leq \delta\}$
$K_{\max}$	Actuator bandwidth limit
$\bar{n}$	Bound on state measurement noise
<i>Closed-Loop RC-ISMC System $\mathcal{H}_{\text{int}}$</i>	
$\xi_e(t)$	System state, $\xi_e(t) = [e_s^\top(t), x_r^\top(t), s(t)]^\top$
A_ξ, E_ξ, B_ξ	Flow matrices of $\mathcal{H}_{\text{int}}$
$A_{\xi,\rho}$	Jump matrix of $\mathcal{H}_{\text{int}}$
$\mathcal{F}_{\text{int}}, \mathcal{J}_{\text{int}}$	Flow and jump sets of $\mathcal{H}_{\text{int}}$
$\mathcal{B}_{\text{reset}}, \mathcal{B}_{\text{ismc}}$	Ultimate tracking-error bounds (reset-only vs. RC-ISMC)
$\mathcal{B}_{\min}$	Ideal-sliding-limit tracking bound ($\delta \rightarrow 0^+$)
$c_q, c_{\text{reset}}, c_{\text{ismc}}$	Constants in the tracking-bound expressions
$v(t)$	Residual matched disturbance under ISMC, $v(t) = d(t) + u_{sm}(t)$
$u_{sm,eq}(t)$	Equivalent sliding mode control effort
ϵ_s	Prescribed ultimate tracking-error tolerance
<i>RC-Λ-ISMC System $\mathcal{H}_\Lambda$</i>	
Λ	Integral-gain matrix
$A_{\xi,\Lambda}, E_{\xi,\Lambda}$	Flow matrices of $\mathcal{H}_\Lambda$
$h_\Lambda^\top$	Coupling row introduced by Λ
$\Psi(t)$	Auxiliary signal isolating ISMC/disturbance terms
$\Psi^*(\rho)$	Conservative envelope bound on $\Psi(t)$
Ψ_{ss}^*	Steady-state (trajectory-dependent) bound on $\Psi(t)$
K_Ψ	Norm constant, $K_\Psi = \ C_m A_\Psi\ $
A_Ψ	Auxiliary matrix used to define $\Psi(t)$
γ_q, γ_v	Scalar gains in the $\Psi^*(\rho)$ envelope
ρ^*	Conservative reachability-guaranteeing gain
$R_\chi(\rho)$	Ultimate bound on $\ \chi(t)\ $
$c_{r,0}$	Flow dissipation constant for the χ -subsystem
$c_{s,0}$	Flow dissipation constant for the s -subsystem
t_Ψ	Time at which steady-state trajectories start
t^*	Times at which Ω_δ is reached
$V(\xi_e)$	Composite Lyapunov candidate, $\chi^\top P \chi + \frac{1}{2}s^2$
$\underline{\alpha}$	$\underline{\alpha} = \min\{\lambda_{\min}(P), 1/2\}$
R_ξ	Ultimate bound on $\ \xi_e(t)\ $

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